



Microprocessor & Computer Architecture (μpCA)

UE24CS251B

Session -
1.1

Course Anchor / s:

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Course

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Introduction

Microprocessor & Computer Architecture

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What we Know? [studied in 3 sem]

Digital Design!

- In DDCO course, *increase logic circuit speed, decrease logic resources required and decrease power consumed.*
- Goal is to Optimize fundamental physical quantities of *time, space and energy.*

Computer

Organization !

- Hardware Abstraction above Digital Logic and below Operating System
- Hardware details which are required by programmers, such as Control signals, Interfaces and Memory

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What we study -

Syllabus

Unit 1: Architecture and Assembly Language Programming

Introduction, ISA Classification - RISC and CISC, Memory Addressing, Operands - Types and Size, Instruction Set - Operations, Control Flow, Instruction Encoding, Case Study - ARM/ MIPS/ x86 Processor.

16

Hours

Unit 2: Pipelining

3- Stage Pipelining, 5 - Stage Pipelining, Pipeline Datapath and Control, Data Hazards – Forwarding vs. Stalling, Control Hazards, Branch Prediction Mechanisms and Exceptions, Performance Metrics.

12 Hours

Unit 3 : Basics of Cache and Cache Optimization

Basics of Caches - Fully Associative, Direct Mapped and Set Associativity, Cache Performance, Basic Cache Optimization- Reduce in Miss Rate. Basic Cache Optimization- Reduce Miss Penalty, Reduce Hit Time.

14 Hours

Unit 4: Advances in Architecture

Introduction to Parallel Computing, Parallel Computer Memory Architecture and Flynn's Taxonomy, Parallel Computing Concepts and Terminology, Design Issues of Operating Systems for multiprocessor systems, Application of

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Evaluation

Policy



	Conduct i on	Scal ed (Marks)
ISA 1 (Units 1 and 2)	40 Marks	20
ISA 2 (Units 3 and 4)	40 Marks	20
Banana Problem	4 Marks	
	Banana problem 1 conducted for 10 marks (Unit 1&2)	
	Banana Problem 2 conducted for 10 marks (Unit 3&4)	
	Each reduced to 2 marks (2*2=4 marks)	10
Orange Problem	6 Marks	
	One Orange problem conducted for 10 marks, reduced to 6 marks	
Laboratory (1-credit) - FSA	- Mini Project component (Jackfruit problem) - 10 marks - Quiz conducted for 20 marks, reduced to 4	

Microprocessor & Computer Architecture (upCA) Text and Reference Book(s)



Text Book(s) :

- T1. "Computer Organization and Design", Patterson, Hennessey, 5th Edition, Morgan Kaufmann, 2014.
- T2. "Computer Organization and Design – ARM Edition", Patterson, Hennessey, 4th Edition, Morgan Kaufmann, 2010.
- T3. "ARM System Developer's Guide Designing and Optimizing System Software", Andrew N Sloss et al., Elsevier, 2004.
- T4. "Programming Massively Parallel Processors A Hands-on Approach", Third Edition David B. Kirk, Wen-mei W. Hwu, Morgan Kaufmann, 2016
- T5. "Multicore and GPU Programming An Integrated Approach", Gerassimos Barlas, second edition, Morgan Kaufmann, 2022

Reference Book(s) :

- R1. "Computer Architecture: A Quantitative Approach", Hennessey, Patterson, 5th Edition, Morgan Kaufmann, 2011.
- R2. "ARM System-on-Chip Architecture", Steve Furber, 2nd Edition, Pearson India, 2015.



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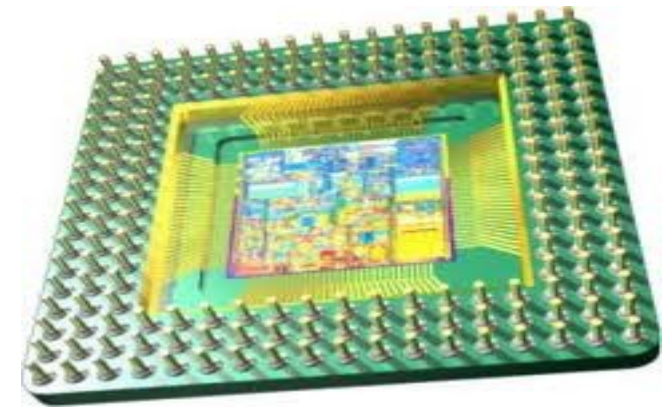
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Introduction to
Microprocessors and
Microcontrollers
RISC and CISC Architectures

Session -
1.11



Is MICROPROCESSOR
Brain or Heart
of a computer System
???



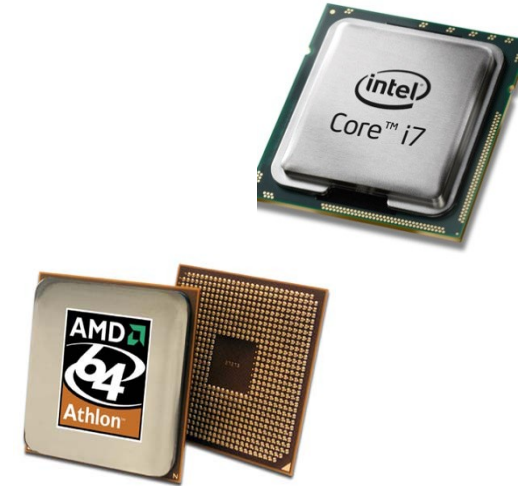
Statutory Warning: does not resemble anyone over here

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What we study?

What is a MicroProcessor?

- Brain of the Computer
- Basic Computing Unit on an Integrated Chip (IC)
- Three basic characteristics differentiate



What is Computer Architecture?

- **Instruction set, Bandwidth, Clock speed** engineering decisions and tradeoffs that must be made in order to produce a **"Good" Design**.

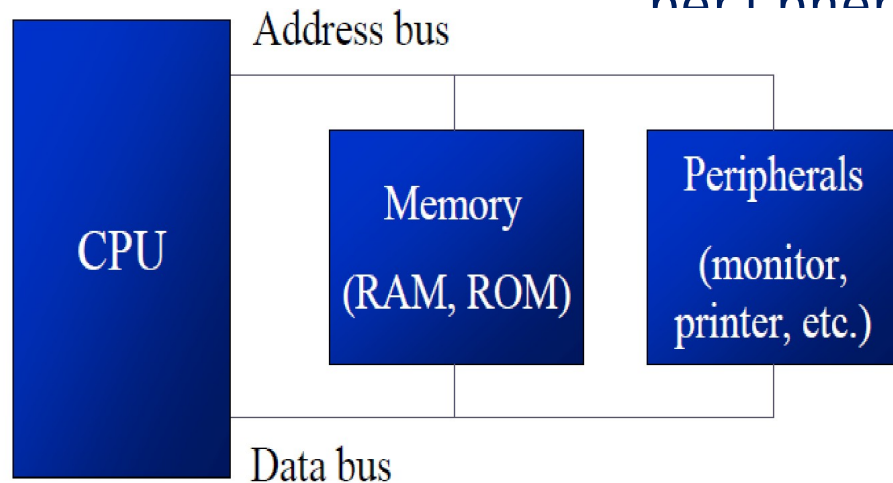
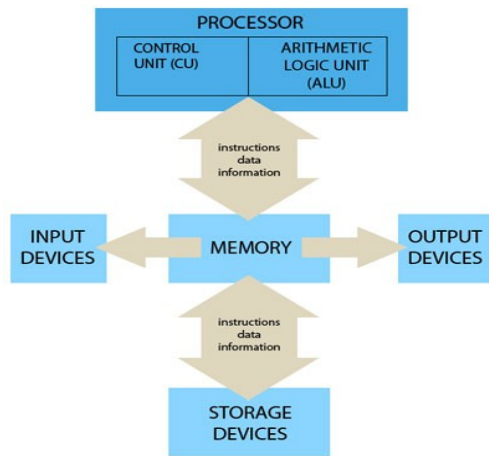
Why Study Microprocessor?

- Everywhere we have devices which are controlled by “Microprocessor” or “Microcontroller”
- Because it's part of the Curriculum!! 😊



Microprocessor & Computer Architecture

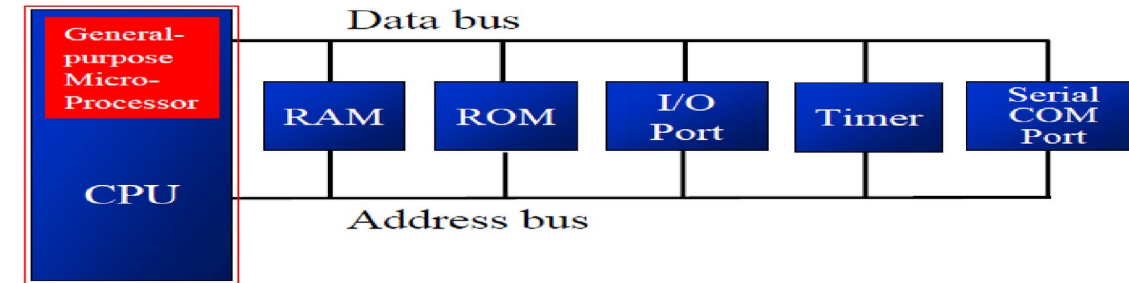
Computer Organization & Computer Architecture



- Computer Organization refers to the **Operational Units and their interconnections** that realize or recognize the specifications of Computer Architecture.
- Organizational **attributes** includes **Hardware details transparent** to the programmer such as **control signals**, **Interface** between computers and peripherals and the memory technology used. **Computer Architecture** refers to those attributes of a system visible to a programmer, the other way those attributes that have direct impact on the logical execution of the program
- Architectural attributes include the **instruction set**, the **no of bits used to represent the data types**, **Input**

Microprocessor & Computer Architecture

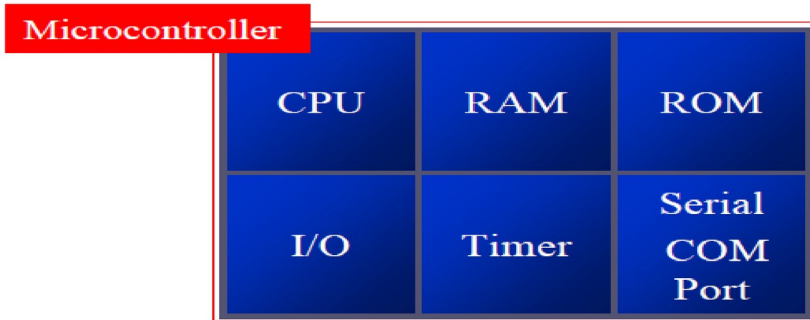
Microprocessor



- Accepts digital data as input, processes it according to instructions stored in its memory, and provides results as output.
- Don't have RAM, ROM, and other peripheral on the chip.
- Memory, I/O and other components need to be connected externally.
- Cannot be used in compact systems.
- Power Consumption is High.

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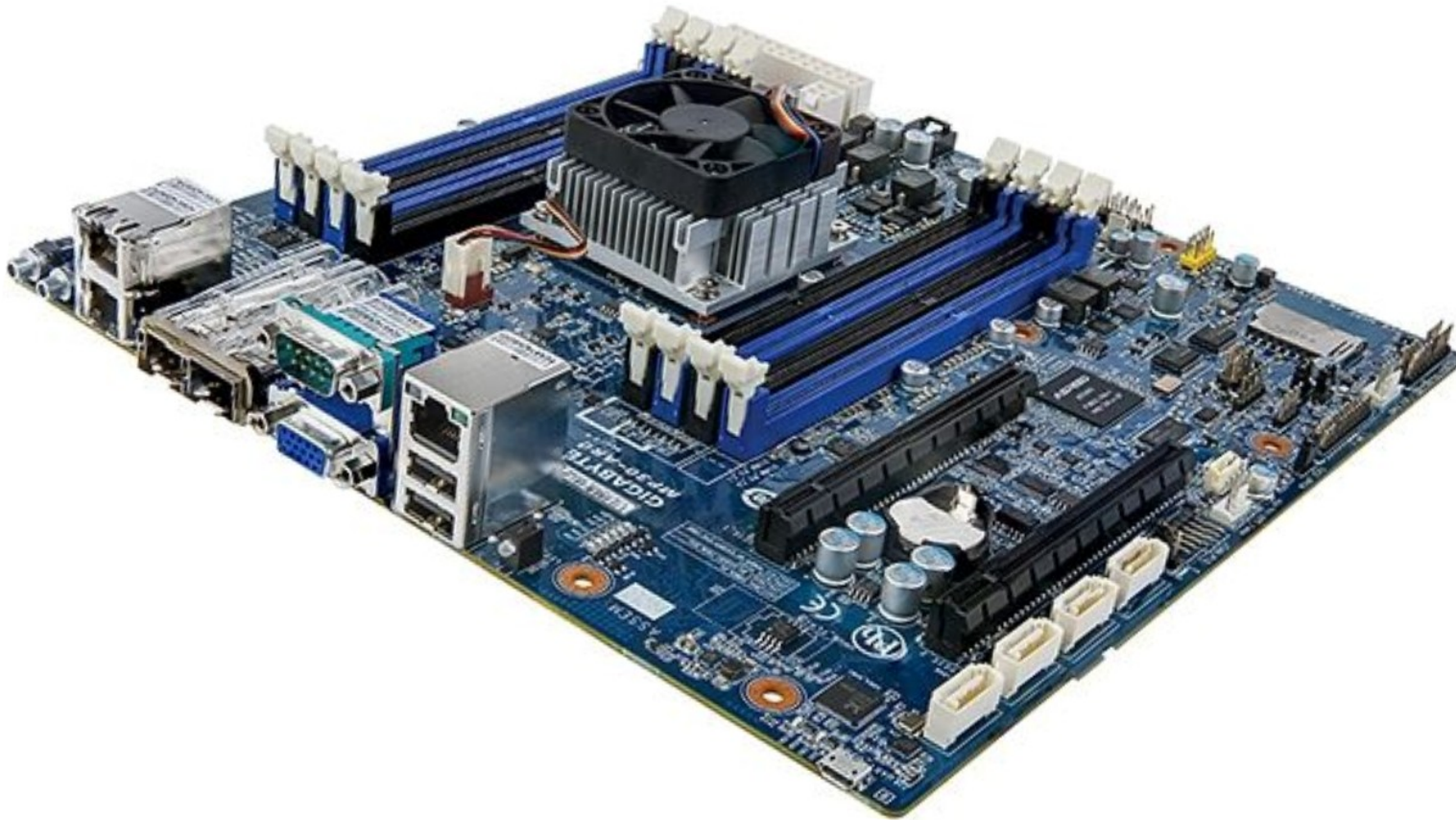
Microcontroller



- Comprises of CPU, in addition with a fixed amount of RAM, ROM and other peripherals all embedded on a single chip.
- Also termed as a mini computer or a computer on a single chip.
- Designed to perform **specific tasks**.
- Requires **small resources** like RAM, ROM, I/O ports etc. and hence can be embedded on a single chip.

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Microprocessor,
Where am I?



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Microprocessor,
Where am I?



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Microprocessor - Evolution

4004	Nov. 15,1971
8008	April 1972
8080	April 1974
8085	March 1976
8086	June 8, 1978
8088	June 1979
80286	Feb. 1982
i80386	1985 - 1990
I80486	1989 - 1992
Intel Pentium	1993 - 1999
Intel Pentium MMX	1996 - 1999
Intel Atom	2008 - 2009 (as Centrino Atom), 2008–present (as Atom)
Intel Celeron	1998–present
Intel Pentium Pro	1995 - 1998
Intel Pentium II	1997 - 1999
Intel Pentium III	1999 - 2003
Intel Xeon	1998–present
Pentium 4	2000 - 2008
Pentium 4	2000 - 2008
Pentium M	2003 - 2008
Pentium D/EE	2005 - 2008

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Microprocessor - Evolution



Intel Pentium Dual-Core	2006 - 2009
Intel Pentium (2009)	2009–present
Intel Core	2006 - 2008
Intel Core 2	2006 - 2011
Intel Core i3	2010–present
Intel Core i5	2009–present
Intel Core i7	2008–present
Intel Core i7 (Extreme Edition)	2011–present
Intel Core i9	2018–present
Intel Core i9 (Extreme Edition)	Q3 2017–present

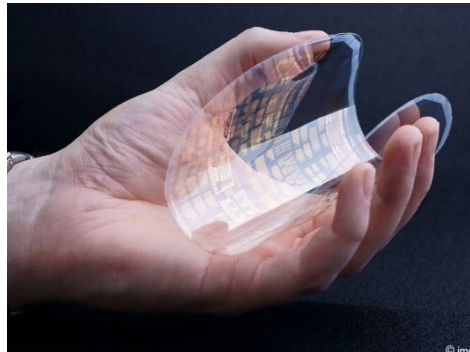
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Why Study

Microprocessor ?

- Because it's part of the Curriculum!! 😊

- Everywhere we have devices which are controlled by “Microprocessor” or “Microcontroller”



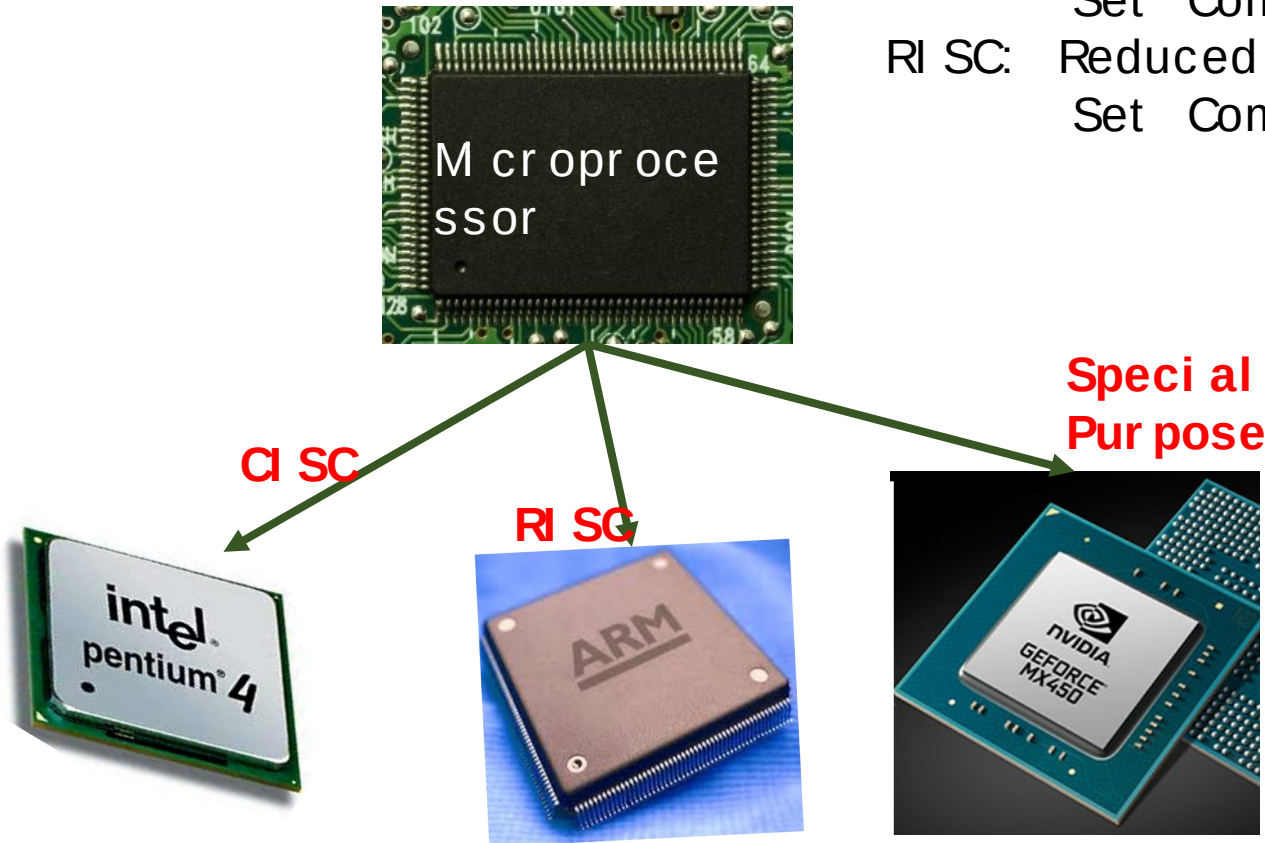
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Classification of

Microprocessors

CI SC: Complex Instruction Set Computer
RI SC: Reduced Instruction Set Computer



How to Classify

Microprocessors?

Three basic characteristics differentiate microprocessors:

- **Instruction set**: The set of instructions that the microprocessor can execute.
- **Bandwidth**: The number of bits processed in a single instruction.
- **Clock Speed**: How many instructions per second the processor can execute.

Given in megahertz (MHz),

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Instruction Set

Architecture (ISA)



- The complete collection of instructions that are understood by a CPU

Example: Data Movement, Data Processing, Branch Instruction...etc

- Designers decide the ISA for respective Microprocessor.
- It is responsibility of Compiler to generate correct and Optimized code for respective Microprocessor.
- Main criteria to classify CISC & RISC Processor

Preamble

- Earliest machines were programmed in assembly language and memory was slow and expensive.
- The CI SC philosophy made sense, and was commonly implemented in such large computers as the PDP-11 and the DEC system 10 and 20 machines.
- Most common microprocessor designs such as the Intel 80x86 and Motorola 68K series followed the CI SC philosophy.
- But recent changes in software and hardware technology have forced a reexamination of CI SC and many modern CI SC processors are hybrids, implementing many RI SC principles and vice versa.
- CI SC was developed to make compiler development simpler. It shifts most of the burden of generating machine instructions to the processor.

- CISC processors were evolved in the 1970s.
- During this period, the computer memory used to be 'small' and 'very expensive'.
- Philosophy of CISC processors is to simplify the code and make it **shorter** in order to reduce the memory requirement.
- In a CISC processor, a single instruction has 'several low-level operations'. This makes the CISC instructions short but 'complex'.

CI SC: Complex Instruction Set Computer

Architecture

- CI SC processors typically had **variable length** instruction sets with **many** formats.
- Typically allowed **values in memory** to be used as **operands** in data processing instructions.
- A **2-operand** instruction format
- **Register sets** were getting **larger**, but **none** was as large as RI SC and most processors had different registers for different purposes.

CI SC Organization

- Microprogrammed control instruction decode logic.
 - **It was easier to implement and less expensive.**
- CI SC processors allowed **little**, if any, overlap between consecutive instructions.
 - **The ease of microcoding new instructions allowed designers to make CI SC machines upwardly compatible**
- May take many clock cycles to complete a single instruction.
 - **Microprogram instruction sets can be written to match the constructs of high-level languages, the compiler does not have to be complicated.**
- Pentium is considered a modern CI SC processor.

RI SC – REDUCED INSTRUCTION SET COMPUTER

- **John Cocke** - (May 30, 1925 – July 16, 2002), was an American computer scientist recognized for his large contribution on computer architecture and optimizing compiler design.
- Originated the **RI SC** concept in **1974**.
- Considered by many to be "**the father of RI SC architecture.**"
- Proved that about 20% of the instructions in a computer did 80% of the work.



teacher at



used in

Silicon Graphics.

concept.

The term **RI SC** is credited to **David Patterson**, a teacher at the University of California in Berkeley. The Sun Microsystems' SPARC microprocessors and led founding of what is now MIPS Technologies, part of A number of current microchips now use the RI SC

- Idea of RLSC processors was originated in the 1974 and implemented in 1980's.
- Philosophy of RLSC processors is to simplify operation of individual instruction.
- Separate instruction for Load and Store (Data Movement)
- **Number of lines of code** in the program **increases** but Amount of work done by individual instruction is **REDUCED**.

RI SC – REDUCED INSTRUCTION SET COMPUTER

Architecture

- A fixed (32-bit).
- A load-store architecture where instructions that process data, operate only on registers and are separate from instructions that access memory.
- A large register bank (thirty-two 32-bit registers).
- Instruction size with few formats.

RI SC Organization

- Hard-wired instruction decode logic (design of the control unit).
- Pipelined execution.
- Single-cycle execution.

RI SC Advantages

- A smaller die size.
- A shorter development time.
- A higher performance.

RI SC Drawbacks

- Poor code density.
- Do not execute x86 code.

RI SC Vs CI SC Machines Today

- With the rise of embedded systems and mobile computing, the terms RI SC and CI SC have lost their significance
- Both architectures almost seem to have adopted the strategies of the other.
- Some RI SC systems provide more extravagant instruction sets than some CI SC systems.
- CI SC chips are now able to execute more than one instruction within a single clock, including Pipelining.
- The RI SC Vs CI SC debate started when chip-area and processor design complexity were issues – now energy and power are the issues.

• The top ten competitors today are ARM and Intel

RI SC – REDUCED INSTRUCTION SET COMPUTER

- State of the art processor technology has changed significantly since RI SC chips were first introduced in the early '80s.
- A number of advancements are used by both RI SC and CI SC processors.
- The lines between the two architectures have begun to blur.
- In fact, the two architectures almost seem to have adopted the strategies of the other.
- Since, the processor speeds have increased, CI SC chips are now able to execute more than one instruction within a single clock.
- This also allows CI SC chips to make use of pipelining.
- With other technological improvements, it is now possible to fit many more transistors on a single chip.

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RISC vs CISC

RISC

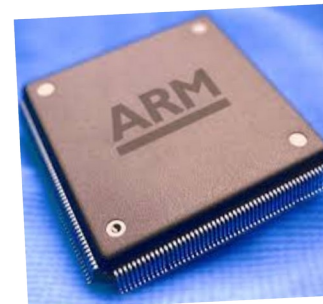
- Simple instructions, few in number.
- Fixed length instructions.
- Multiple register sets.
- Three operands per instruction.
- Parameter passing through register windows.
- Single-cycle instructions.
- Hardwired control.
- Highly pipelined.
- Complexity in compiler.
- Only **LOAD/STORE** instructions access memory.
- Few addressing modes.

CISC

- Many complex instructions.
- Variable length instructions.
- Single register set.
- One or two register operands per instruction.
- Parameter passing through memory.
- Multiple cycle instructions.
- Microprogrammed control.
- Less pipelined.
- Many instructions can access memory.
- Many addressing modes.

Microprocessor & Computer Architecture

Our choice -> ARM



Most likely
on- - - - -



THANK YOU



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